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Enhancement of Antimicrobial Peptide Activity by Ribonuclease

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Abstract

Background: HNP1, LL-37, and HBD1 are antimicrobial against *Escherichia coli* ATCC 25922 at the standard inoculum but less active at higher inocula.
Methods: The virtual colony count (VCC) microbiological assay was adapted for high inocula and the addition of yeast tRNA and bovine pancreatic ribonuclease (RNase). 96-well plates were read for 12 hours in a Tecan Infinite M1000 plate reader and photographed under 10x magnification.
Results: Adding tRNA 1:1 to HNP1 at the standard inoculum almost completely abrogated activity. Adding RNase 1:1 to HNP1 at the standard inoculum of $5x10^5$ CFU/mL did not enhance activity. Increasing the inoculum to $6.25x10^7$ CFU/mL almost abrogated HNP1 activity. However, adding RNase 25:1 to HNP1 enhanced activity. Adding both tRNA and RNase resulted in enhanced activity, indicating that the enhancement effect of RNase overwhelms the inhibiting effect of tRNA when both are present. HBD1 activity at the standard inoculum was almost completely abrogated by the addition of tRNA, but LL-37 activity was only slightly inhibited by tRNA. At the high inoculum, LL-37 activity was enhanced by RNase. HBD1 activity was not enhanced by RNase. RNase was not antimicrobial in the absence of antimicrobial peptides. Cell clumps were observed at the high inoculum in the presence of all three antimicrobial peptides and at the standard inoculum in the presence of HNP1+tRNA.
Conclusions: Antimicrobial peptide-ribonuclease combinations have the potential to be active against high cell concentrations and biofilms, conditions where the antimicrobial agent alone is relatively ineffective.

Introduction

Cationic antimicrobial peptides (CAPs) such as the human defensins and LL-37 kill bacteria potently when concentrations of bacteria are kept low. Previously, I demonstrated a pronounced inoculum effect when the human alpha defensin HNP-1 was assayed against *Escherichia coli*, *Staphylococcus aureus*, and *Bacillus cereus*.[1] However, the molecular basis of the inoculum effect remains unclear. Here I hypothesize that the polyanion tRNA might inhibit CAPs, that inhibition by RNA is partially responsible for the inoculum effect, and that the addition of ribonuclease could enhance antimicrobial peptide activity.

Methods

The VCC assay was adapted for high inocula as previously described[1], and volumes were adjusted to allow for the addition of yeast tRNA and/or bovine pancreatic RNase. 96-well plates were read for 12 hours in a Tecan Infinite M1000 plate reader and photographed under 10x magnification. In some experiments, the concentration of tryptic soy broth (TSB) present in phosphate buffer was adjusted.

Results

Adding tRNA 1:1 to HNP1 at the standard inoculum almost completely abrogated activity (Figure 1). Adding Roche RNase 1:1 to HNP1 at the standard inoculum of $5x10^5$ CFU/mL did not enhance activity. Increasing the inoculum to $6.25x10^7$ CFU/mL almost abrogated HNP1 activity. (Figure 2) However, adding RNase 25:1 to HNP1 enhanced activity at the high inoculum. Adding both tRNA and RNase at the high inoculum resulted in enhanced activity, indicating that the enhancement effect of RNase overwhelms the inhibiting effect of tRNA when both are present. HBD1 activity at the standard inoculum was almost completely abrogated by the addition of tRNA, but LL-37 activity at the standard inoculum was only slightly inhibited by tRNA. (Figure 3) At the high inoculum, LL-37 activity was enhanced by RNase, but LL-37 showed greater activity than HNP1 in the absence of RNase. (Figure 4) HBD1 activity was not enhanced by RNase. RNase was not antimicrobial at the high inoculum without antimicrobial peptides. The observations with HNP1 at the high inolculum were repeated using a second RNase manufacturer, Macherey-Nagel (MN). (Figure 5) The experiment with MN RNase was repeated. (Figure 6) 1% TSB was used in most assays, but the %TSB was varied in one experiment, resulting in maximum activity at 4% TSB with either 5x or 25x MN RNase added. (Figure 7) Cell clumps were observed at the high inoculum in the presence of all three antimicrobial peptides

Results

Figure 1: HNP1 vs. Standard Inoculum

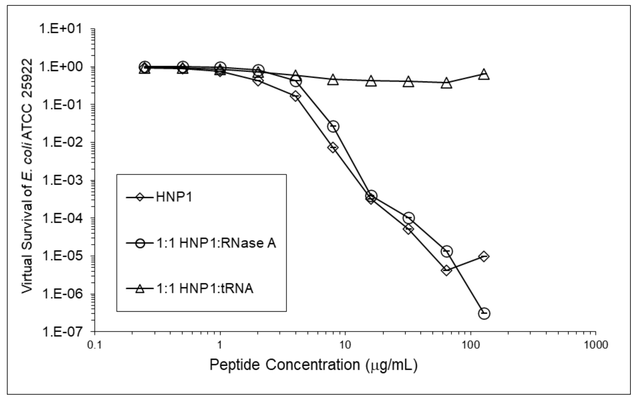


Figure 2: HNP1 vs. High Inoculum

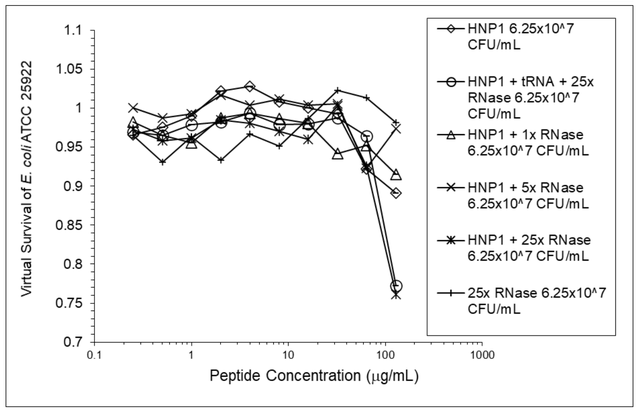
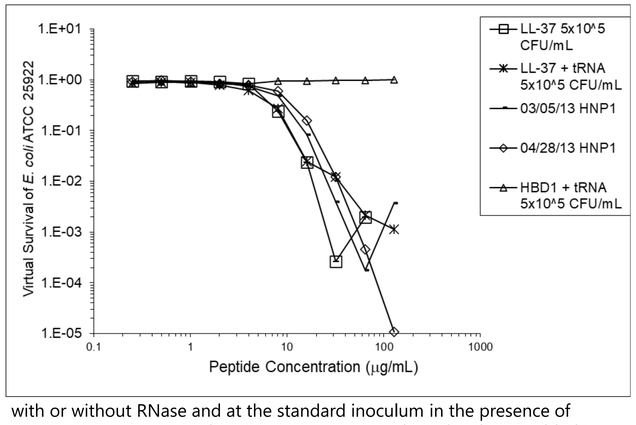


Figure 3: HNP1, HBD1 and LL-37 vs. Standard Inoculum



Results

Figure 4: HBD1 and LL-37 vs. High Inoculum

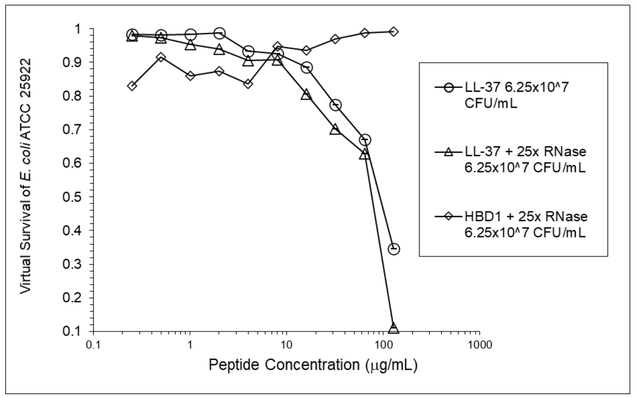


Figure 5: HNP1 vs. High Inoculum, Second RNase Manufacturer

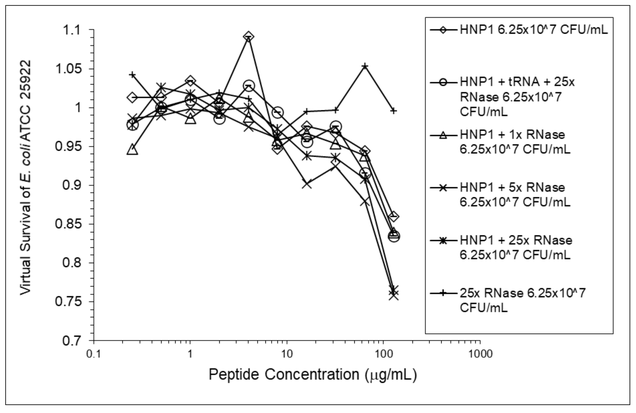
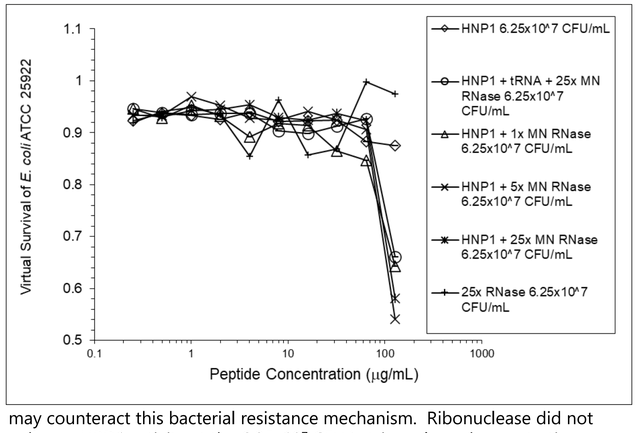


Figure 6: HNP1 vs. High Inoculum, Second Replicate



with or without RNase and at the standard inoculum in the presence of HNP1+tRNA. (Figure 8) The VCC assays were conducted with TSB added to the 10 mM sodium phosphate incubation buffer. The enhancement of activity caused by RNase was observed with LL-37 but not HNP1 when washed cells were used, indicating that RNase operates by different mechanisms with the two antimicrobial peptides. Although biofilm formation was not directly assayed, it is assumed that the cell clumps photographed at 10x magnification are biofilms. Therefore, lower activity at the high inoculum may be a consequence of biofilm formation, and the enhancement effect of RNase may counteract this bacterial resistance mechanism. Ribonuclease did not enhance HBD1 activity at the $6.25x10^7$ CFU/mL inoculum, demonstrating a strong inoculum effect with HBD1 vs. *E. coli*. LL-37 had a lesser inoculum effect against *E. coli*. The effect of ribonuclease on HNP1 is strongest with lowest amounts of TSB present in the phosphate buffer during the 2 hour incubation. The ability of tRNA to abrogate HNP1 and HBD1 activity, and the failure of tRNA to abrogate LL-37 activity, at the standard inoculum cannot be explained by net charge, indicating hydrophobic interactions also play a role, or that inhibition by tRNA works indirectly through biofilm formation.

Results

Figure 7: HNP1 vs. High Inoculum, Variation in %TSB

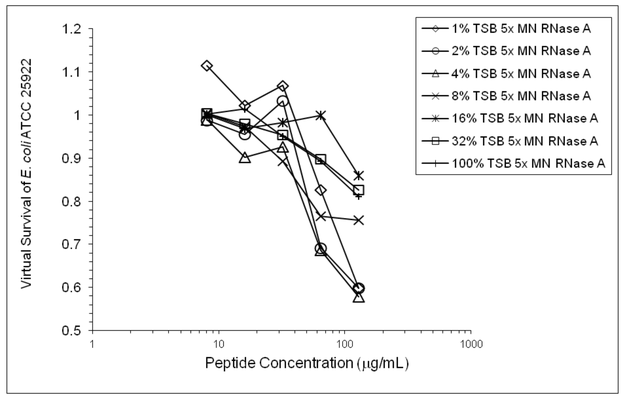


Figure 8: Cell Clumps

Well B2, 128 μg/mL HNP1 vs. $6.25x10^7$ CFU/mL *E. coli*

Well C2, 128 μg/mL HNP1 + 1:5 MN RNase vs. $6.25x10^7$ CFU/mL *E. coli*

Conclusions

The inhibitory presence of extracellular RNA would be expected to increase with increasing cell concentrations. Some human ribonucleases, such as RNase 1 and RNase 7, are potent antimicrobial peptides. However, bovine pancreatic ribonuclease has not previously been shown to be antimicrobial. Both RNA and RNases are ubiquitous in vivo. Therefore, these experiments may be more biologically relevant than VCC experiments lacking RNA or RNase. They offer an insight into activity that may be important for host defense. LL-37 and RNase 1 have been shown to act synergistically to kill *E. coli*.[2] RNases have been tested in clinical trials as chemotherapeutics for the treatment of cancer.[3] A new generation of antimicrobial peptide-RNase combinations offer a new hope that peptides that that are sometimes defeated by the resistance mechanism of biofilm formation can be repurposed to degrade biofilms instead, with increased activity to fight acute infections.

Acknowledgments

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References

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